Nishant **Singh**

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Education

Indian Institute of Information Technology Bhopal *Bhopal, MP*

B.Tech in Information Technology | GPA: 8.73 *Dec, 2021 – Jun, 2025*

Relevant Coursework: Data Structures and Algorithm, Database Management System, Operating System, Computer Network

Work Experience

Ultimate Kronos Group *Pune, MH*

Software Engineer *Jul, 2025 – Present*

- Owned and delivered **Roll-Up View** for salary planning platform (Angular, Spring Boot, PostgreSQL), enabling real-time compensation aggregation and multi-currency normalization at **12.8M employee** scale; supported April 2026 release with stake of **$7.5M** ARR
- Reduced pay scale update time from **10 mins** to seconds, eliminating **365**+ hours of manual effort per cycle for **640** enterprise users
- Built and productionized GCP infrastructure integrating **7**+ distributed systems (Kafka, Redis, Denodo), provisioning multi-environment deployments and executing DB schema migrations; enabled first production rollout of a **$7.5M** ARR platform at **20K-employee** scale
- Remediated a high-severity (**CVSS 8.1**) stored **XSS** vulnerability by implementing **OWASP**-aligned input sanitization and output encoding, preventing session hijacking and multi-user compromise risk across **1,600**+ enterprises and **12.8M** employee records

Google Summer of Code *Remote*

Mentor *Mar, 2025 – Nov, 2025*

- Mentored contributors to design & implement a scalable observability layer for KubeArmor, integrating **6+ Prometheus** metrics to expose real-time insights into policy enforcement, alerting, & overall security activity across Kubernetes & non-Kubernetes workloads
- Introduced and exposed **4+** key production-grade metrics endpoints in **Go**, adhering to Prometheus exposition standards and enabling seamless integration with observability tools like **Grafana**, Alertmanager, & custom dashboards, leading to improved **observability**

Samsung Research Institute - Delhi *Noida, UP*

Software Developer Intern *Jan, 2025 – May, 2025*

- Implemented **runtime** API validation using the validator package in **Go**, adding field-level constraints like minimum/maximum values to Swagger specifications for **11 Iris-based** backend APIs, reducing **80%** of validation errors and improving overall API reliability
- Revamped UMD movie matching by manually mapping unmatched metadata entries between **Samsung**'s UMD and IMDb database, and developing a custom CLI tool that reduced the task time from **4–5 hours** to just **30 minutes** daily

GoComet Pvt Ltd *Bangalore, KA*

DevOps Engineer Intern *Oct, 2023 – Oct, 2024*

- Established a robust monitoring infrastructure by configuring alerts for **PostgreSQL 13**, specifically targeting the long idling transactions, locks and max concurrent connections, providing immediate visibility into potential issues and enabling swift resolution
- Spearheaded the daily deployment process of production code through a enhanced rolling deployment strategy on Kubernetes, orchestrating seamless updates across **5+ systems** and leveraging automated tools to minimize downtime & enhance system reliability
- Enhanced operational insights by setting up **Fluentd logging** agent to push detailed, enriched application logs to **New Relic**, enabling deeper understanding of application behavior, streamlined troubleshooting the intricacies, reducing average debugging time **by 30%**

Projects

Blaze *tesla59/blazetv*

Go, NextJS, PostgreSQL, WebSocket, WebRTC

- Coded a scalable Omeagle-like real-time video chat platform using Gorilla **WebSockets** in **Go**, achieving **10,000+** concurrent WebSocket connections on a single core, structured using SOLID principles and a clean Repository-Service-Controller architecture
- Leveraged PostgreSQL to persist **100,000**+ user sessions, metadata, and abuse reports, while building a **Next.js** frontend integrated with **WebRTC**, supporting **5,000**+ simultaneous peer-to-peer video/audio streams, coordinated through a custom Go signaling server
- Transformed & organised robust **CI/CD** pipelines using GitHub Actions, automating deployment workflows reducing release times by **70**%, and containerized the application with Docker to enable consistent and scalable deployments across multiple environments

GoLoadBalancer *tesla59/GoLoadBalancer*

Go, Docker, Load Balancer, Concurrency

- Developed an automatic load balancer in Go **v1.21** with round-robin functionality, inspired by Kubernetes **v1.24** operator architecture
- Leveraged Goroutine and Docker to spawn **3**+ load balancer to balance the incoming load and 5+ workers to handle incoming request
- Incorporated **SQLite 3** Database to store and provide real-time statistics such as of all workers such as average failure rate and delay

WhisperWeb *tesla59/whisperweb-backend*

Go, Fiber, REST APIs, SQLite, Docker

- Coded a HTTP Server using Fiber web framework in **Go** for back-end using **SQLite 3** as the DBMS to store and retrieve data
- Designed the front-end part using **Next JS 13** capitalising Server-Side Rendering (SSR) pattern to enhance performance and experience
- Containerized the website using **Docker** ensuring portability and reliability across **5**+ different environments and easy deployment

Skills

Languages: Go, Java, C, C++, Rust, TypeScript, JavaScript, Dart, Python
Tools: Docker, Kubernetes, Terraform, Ansible, Git, ArgoCD, SQL, Jenkins
Frameworks: SpringBoot, Selenium, Iris, Gorilla, Flutter, NextJS, GoFiber, Gorm
Platforms: Linux, AWS Cloud, Google Cloud Platform, Azure, Oracle Cloud

Certificates

Jan, 2024 **Google Cloud Certified Professional Cloud Architect**, *Google Cloud*
Nov, 2023 **Google Cloud Certified Associate Cloud Engineer**, *Google Cloud*